

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – X
PAPER –2
TEST DATE: 21-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

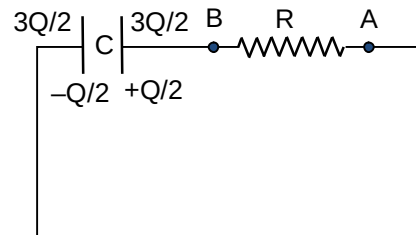
Section – A

1. B, C

Sol. $Q(t) = \frac{Q}{2} (e^{-t/RC})$

$$I(t) = \frac{Q}{2RC} e^{-t/RC} \quad (\text{from B to A})$$

$$\text{Total heat dissipated} = \frac{(Q/2)^2}{2C} = \frac{Q^2}{8C}$$



2. A, D

Sol. $\left(\frac{D+x}{D-x} \right)^2 = \frac{9}{1}$

$$\frac{D+x}{D-x} = 3$$

$$D+x = 3D-3x$$

$$4x = 2D$$

$$x = \frac{D}{2} = \frac{120}{2} = 60 \text{ cm}$$

$$x = 60 \text{ cm}$$

$$\text{Now, } f = \frac{D^2 - x^2}{4D} = \frac{(120 \times 120) - (60 \times 60)}{4 \times 120}$$

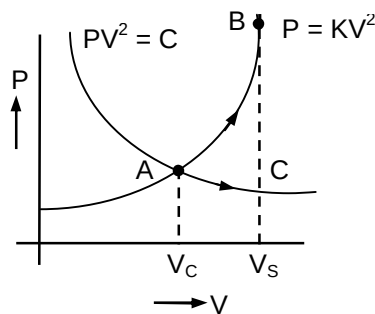
$$f = 22.5 \text{ cm}$$

3. B, C

Sol. The two springs have different spring constants. Also energy remains conserved during the motion as no friction is present.

4. B, D

Sol. From equation, $PV = nRT$
 $P_C < P_B$, $V_C = V_B$ and $T_B > T_C$
 $W_{AB} > W_{AC}$; $(T_B - T_A) > (T_C - T_A)$
 So, by 1st law, $Q = \Delta U + W$
 $Q_{AB} > Q_{AC}$.



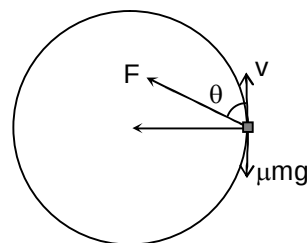
5. A

Sol. $F \cos \theta = \mu mg$

$$F \sin \theta = \frac{mv^2}{R}$$

$$F = \sqrt{(\mu mg)^2 + \left(\frac{mv^2}{R}\right)^2}$$

$$F = \sqrt{2} \text{ N.}$$



6. B, D

Sol. **Case I:**

Since there is no relative motion,

$$a = \frac{F_1 - F_f}{5} = \frac{F_f}{3} \rightarrow F_{1(\max)} = \frac{8}{3} F_f$$

Case II:

$$a = \frac{F_f}{5} = \frac{F_2 - F_f}{3} \Rightarrow F_{2(\max)} = \frac{8}{5} F_f$$

Clearly, $F_{1(\max)} > F_{2(\max)}$ and $\frac{F_{1(\max)}}{F_{2(\max)}} = \frac{5}{3}$.

7. D

Sol. $\lambda = \frac{v}{n} = \frac{332}{1000} = 0.332 \text{ m.}$

8. C

Sol. The apparent frequency for reflecting surface as observer is

$$n' = 1000 \left[\frac{332 + 64}{332 - 32} \right] = 1320 \text{ Hz}$$

9. C

10. B

Sol. **(for Q. 9 to 10):**

The maximum acceleration with which the blocks can move together without slipping

$$a_{\max} = \frac{\mu mg}{3M} = \frac{0.3 \times 1 \times 10}{3 \times 2} = 0.5 \text{ m/s}^2$$

$$F_{\max} = 2(M + m)a_{\max} = 6 \times 0.5 = 3 \text{ N}$$

The friction force between the front blocks,

$$F_s = (M + 2m)a_{\max} = 4 \times 0.5 = 2 \text{ N}$$

Section – B

11. 4

Sol. $a + R\alpha = A\omega^2 \cos \omega t$... (i)

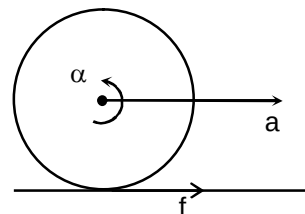
$f = ma$... (ii)

$fR = \frac{1}{2}mR^2\alpha$... (iii)

$f + 2f = mA\omega^2 \cos \omega t$

$f = \frac{1}{3}mA\omega^2 \cos \omega t$

$\tau_{(\max)} = \frac{1}{3}mA\omega^2 R = \frac{1}{3} \times 6 \times 2 \times 1 = 4 \text{ N-m}$

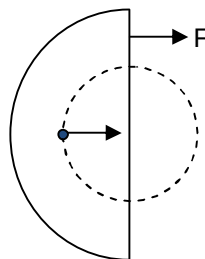


12. 20

Sol. $\Delta \ell_{\max} = \frac{2}{k} \left(\frac{m_1 F_2 + m_2 F_1}{m_1 + m_2} \right) = \frac{2}{150} \left(\frac{2 \times 27 + 4 \times 9}{2 + 4} \right) = \frac{2}{150} \times \frac{90}{6} = 0.20 \text{ m} = 20 \text{ cm}$

13. 3

Sol. $F = \left(\frac{m}{2} \right) \omega^2 \frac{3r}{8} = \frac{3mv^2}{16r}$



Section – C

14. 03.40

15. 04.74

Sol. (for Q. 14 to 15):

Using conservation of momentum

$P = P_2 \cos \theta$... (i)

$P_1 = P_2 \sin \theta$... (ii)

Squaring and adding, we get

$P^2 + P_1^2 = P_2^2$

$2mk + 2m_1 k_1 = 2m_2 K_2$

$4k + k_1 = 10k_2$

$10k_2 - k_1 = 4k$

$\therefore 10k_2 - k_1 = 44$... (iii)

Also, $k_1 + k_2 = k - |Q|$

$k_1 + k_2 = 11 - 2.86$

$k_1 + k_2 = 8.14$... (iv)

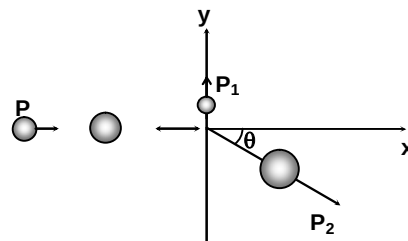
Adding (iii) and (iv), we get

$11k_2 = 52.14$

$\therefore k_2 = 4.74 \text{ MeV}$

$\therefore k_1 = 8.14 - 4.74$

$k_1 = 3.40 \text{ MeV}$



16. 03.00

17. 01.00

Sol. **(for Q. 16 to 17):**

$$4Ta_B - 3Ta_A = 0 \Rightarrow a_B = \frac{3a}{4} \Rightarrow M = 3$$

$$2Ta_B - Ta_A - Ta_P = 0 \Rightarrow a_P = 2a_B - a_A = \frac{3a}{2} - a = \frac{a}{2}$$

$$a_P = \frac{a}{2} \Rightarrow K = 1$$

18. 180.00

19. 02.00

Sol. **(for Q. 18 to 19):**

At the instant shown both particles are at their mean positions and moving in opposite direction.

Phase difference = 180°

As ω is same for both the particles

$\left\{ \omega = \sqrt{\frac{GM}{R^3}} \right\}$ the phase difference will be maintained throughout and they can never meet.

$v_{\max} = \omega A \rightarrow$ For a particle undergoing S.H.M

$$\text{So, } \frac{v_1}{v_2} = \frac{R}{R/2} = \frac{2}{1}$$

Chemistry**PART – II****Section – A**

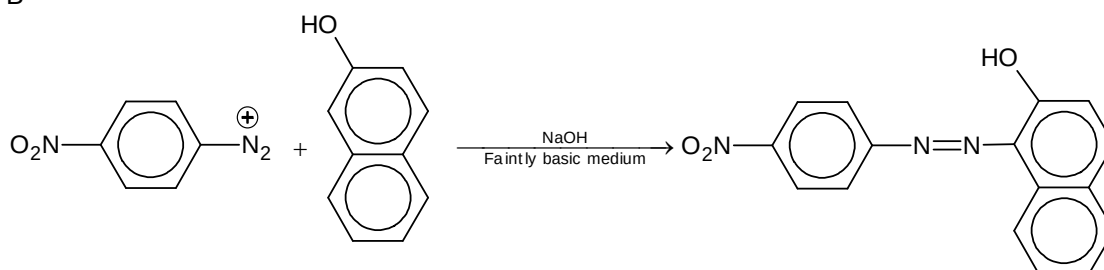
20. B, C

Sol. The V.D. and molar mass of the emergent gas will always be greater than that of pure gas.

- In case of CH_3OH (molar mass 32 g mol^{-1}) and acetone (molar mass 58 g mol^{-1})
- Molar mass of N_2 being 28 g mol^{-1}
- Molar mass of H_2O being 18 g mol^{-1} .
- Molar mass of D_2O being 20 g mol^{-1} .

21. B

Sol.



22. A, B, C

- Sol. (A) Below critical micelle concentration soap behaves like normal electrolyte; no micellisation of the anions.
- (B) In water-in-oil emulsion, oil forms external (continuous) phase.
- (C) $\text{SnO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SnO}_3 + \text{H}_2\text{O}$
- (D) Cations will be effective in coagulation of negatively charged sol formed in option (C).

23. A, B, D

- Sol. 1. S(entropy) is not the measure of available energy.
2. ΔS between the two states is not path dependent whether reversible or irreversible $\Delta S = S_2 - S_1$.
3. Heat supplied into the isothermal reversible expansion of ideal gas is equal to work of expansion of the gas.
4. $-\Delta G$ is equal to the available energy of the system to do useful work.

24. A, B, C, D

- Sol. 1. Ethers have much lower boiling point than alcohol of similar molecular mass, alcohols undergo molecular association through intermolecular hydrogen bonding.
2. Intramolecular hydrogen bonding makes ortho-isomer weaker acid than para-isomer.
3. At lower temperature (20°C), the reaction between phenol and conc. H_2SO_4 is kinetically controlled and product composition depends on relative rate of formation.
4. Owing to intramolecular H-bonding in ortho-nitrophenol, its boiling point is less than that of p-nitrophenol (molecular association due to inter molecular H-bonding).

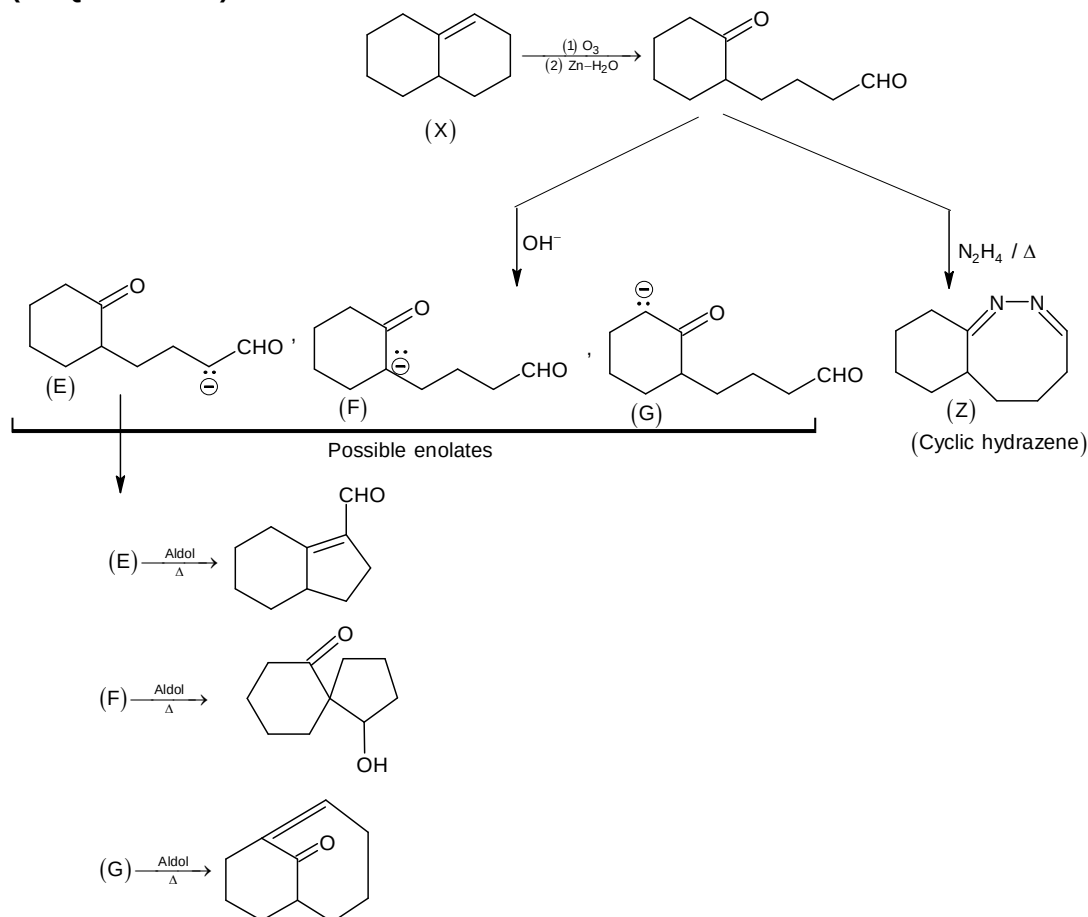
25. A, B, C

- Sol. Brass – $\text{Cu} + \text{Zn}$
 Bell metal – $\text{Cu} + \text{Sn}$
 Nickeloy – $\text{Al} + \text{Cu} + \text{Ni}$
 Magnalium – $\text{Al} + \text{Mg} + \text{Ni}$

26. D

27. B

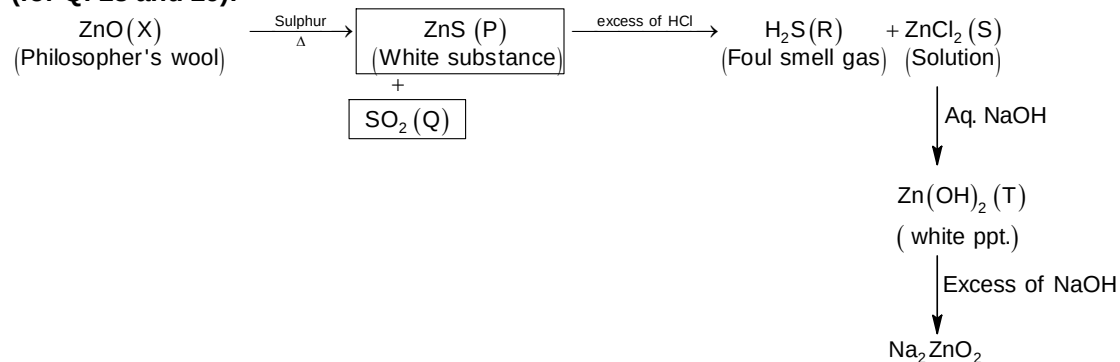
Sol. (for Q. 26 and 27):



28. B

29. D

Sol. (for Q. 28 and 29):



Sol. 28. (A)

ZnO – is an amphoteric oxide.

(B)

ZnO is white powder which on heating turns yellow but again on cooling becomes white.

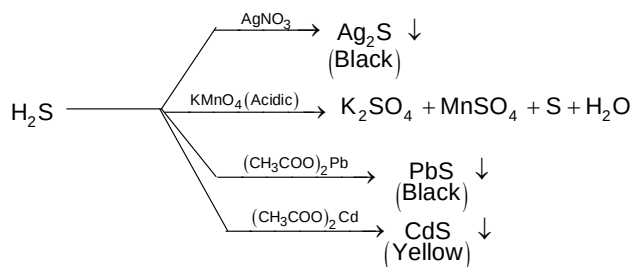
(C)

 Lithopone is mixture of $\text{ZnS} + \text{BaSO}_4$

(D)

 $\text{ZnO} + \text{CoO} \longrightarrow \text{CoZnO}_2 \text{ or } \text{CoO} \cdot \text{ZnO}$
 Rinmann's green

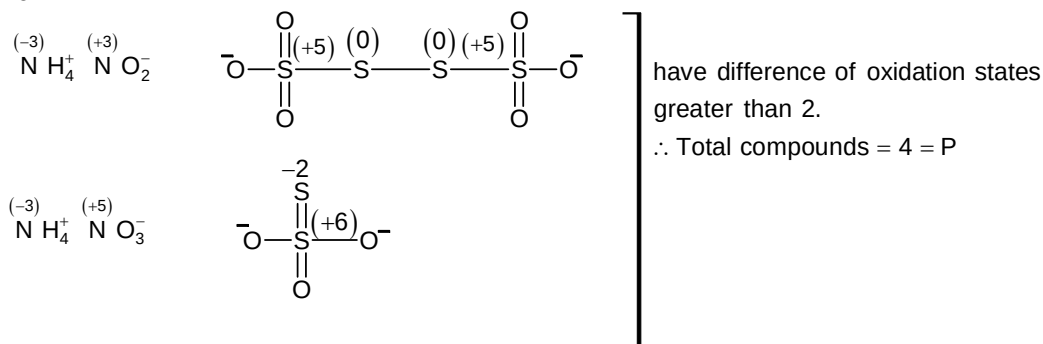
Sol. 29.



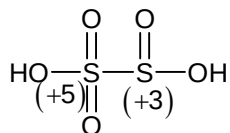
Section - B

30. 16

Sol.



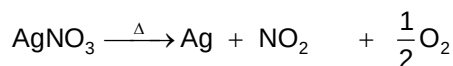
- P
- Hydrazine (NH_2NH_2) – [Both nitrogen have same oxidation state of (-2)]
 - Calcium cyanamide (CaNCN) – [Both nitrogen have same oxidation state of (-3)]
 - Urea (NH_2CONH_2) – [Both nitrogen have same oxidation state of (-3)]
 - $[\text{Cr}_2\text{O}_7]^{2-}$ both chromium have same oxidation state = 6



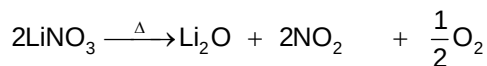
$$\therefore P \times Q = 4 \times 4 = 16$$

31. 92

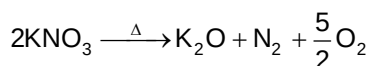
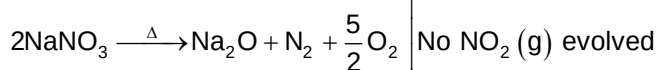
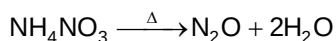
Sol.



(0.5 mole) (0.5 mole)



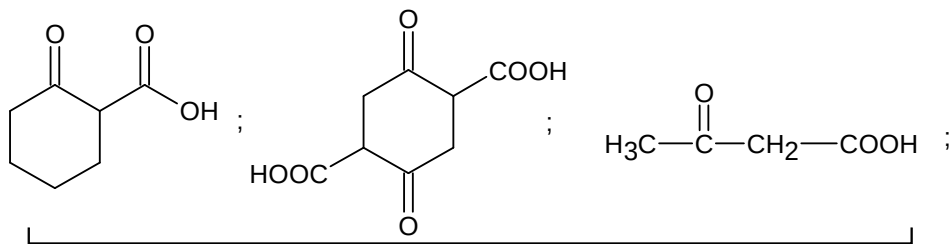
(0.5 mole) (0.5 mole)

(0.5 mole) $0.5 \times 2 = (1 \text{ mole})$ No NO_2 (g) evolved

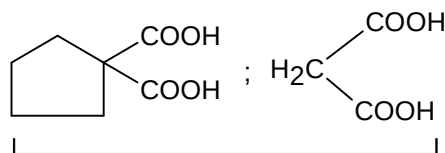
∴ Total number of moles evolved $0.5 + 0.5 + 1 = 2$ mole

Total weight = 2×46 (mol. mass of NO_2) = 92 g

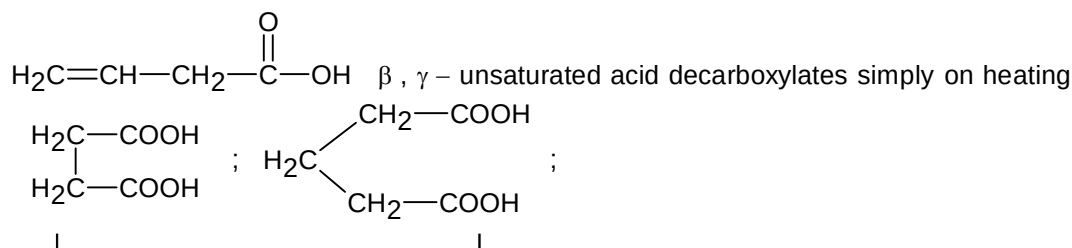
32.
Sol.



β - keto acid decarboxylates on simple heating



Geminal diacids also decarboxylates on heating



1,4 and 1,5 - diacid dehydrates on heating.

Section - C

33. 00.10

34. 04.00

Sol. (for Q. 33 and 34):

Partial pressure of X_2 before reaction $p_{\text{X}_2} = \frac{1 \times 2}{2 + 3} = 0.4$ atm

Partial pressure of Y_2 before reaction $p_{\text{Y}_2} = \frac{2 \times 3}{2 + 3} = 1.2$ atm

Atm of X_2 (g) and Y_2 (g) each reacted = $\frac{0.6}{2} = 0.3$ atm

p_{X_2} (at equilibrium) = $0.4 - 0.3 = 0.1$ atm

p_{Y_2} (at equilibrium) = $1.2 - 0.3 = 0.9$ atm

$$K_p = K_c = \frac{p_{\text{XY}}^2}{p_{\text{X}_2} \times p_{\text{Y}_2}} = \frac{(0.6)^2}{0.1 \times 0.9} = 4.0$$

35. 02.00

36. 11.00

Sol. (for Q. 35 and 36):

$$[\text{CN}^-] \text{ in the resulting solution} = \frac{2.0 \times 0.05 + 3.0 \times (2 \times 0.05)}{2 + 3}$$

$$= \frac{0.4}{5} = 0.08 \text{ M}$$

$$K_w = [\text{H}^+][\text{OH}^-] = 2 \times 10^{-7} \times 2 \times 10^{-7} \\ = 4.0 \times 10^{-14} \text{ M}^2$$



$$K_a (\text{of HCN}) = \frac{K_w}{K_b} = \frac{4 \times 10^{-14}}{2 \times 10^{-4}} = 2 \times 10^{-10}$$

$$\text{'X'} - \text{Hydrolysis constant } K_H \text{ of } \text{CN}^- = \frac{K_w}{K_a} = \frac{4 \times 10^{-14}}{2 \times 10^{-10}} = 2.0 \times 10^{-4}$$

$$\begin{aligned} \text{'Y'} - \text{pH} &= \frac{1}{2} [\text{p}K_w + \text{p}K_a + \log C] \\ &= \frac{1}{2} [-\log 4 \times 10^{-14} - \log 2 \times 10^{-10} + \log 0.08] \\ &= \frac{1}{2} [14 - 2\log 2 + 10 - \log 2 + \log(2)^3 \times (10^{-2})] \\ &= \frac{1}{2} [24 - 3\log 2 - 2 + 3\log 2] \\ &= 11 \end{aligned}$$

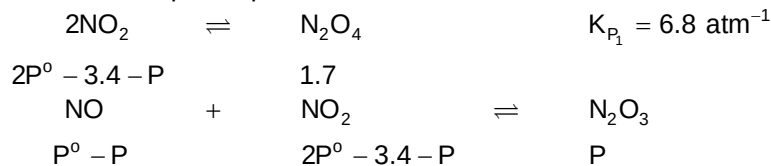
37. 01.05

38. 03.43
(3.42 – 3.43)

Sol. (for Q. 37 and 38):

Let P^0 be initial partial pressure of NO.

$2P^0$ be initial partial pressure of NO_2 .



$$K_P (\text{for 1st equilibrium}) \quad 6.8 = \frac{1.7}{(2P^0 - 3.4 - P)^2}$$

$$\text{or, } 2P^0 - 3.4 - P = 0.5$$

$$\text{or, } 2P^0 - P = 3.9 \quad \dots (1)$$

Also, total pressure (at equilibrium) $P_{\text{NO}_2} + P_{\text{N}_2\text{O}_4} + P_{\text{NO}} + P_{\text{N}_2\text{O}_3}$

$$5.05 = 0.5 + 1.7 + P^0 - P + P$$

$$P^0 = 2.85 \text{ atm}$$

Now from Equation (1) $2P^0 - P = 3.9$

$$2 \times 2.85 - P = 3.9 \Rightarrow P = 1.8 \text{ atm}$$

$\therefore$ Equilibrium pressure of NO = $P^0 - P = 2.85 - 1.80 = 1.05 \text{ atm}$.

$$K_{P_2} = \frac{P_{\text{N}_2\text{O}_3}}{P_{\text{NO}} \times P_{\text{NO}_2}} = \frac{1.8}{(1.05)(0.5)} = 3.42 \text{ atm}^{-1}.$$

Mathematics

PART – III

Section – A

39. A, D

Sol. $\sum_{a \in B} |a| = 666, \sum_{a \in C} |a| = \sum_{a \in D} |a| = 0$

40. A, B, C

Sol. $C : x^2 + y^2 = b^2$
 $S : x^2 + y^2 = a^2$ where $a^2 = 2b^2$

41. A, B, C, D

Sol. if $k = 0 \Rightarrow P = 0$ and if $k = r \Rightarrow P = 2022!$

42. A, C

Sol. Area of R = $\frac{4}{2/3} = 6$

Equation of R : $x + 2y + (2 + \lambda)z - 30 - 15\lambda = 0$ where $\frac{2}{3} = \frac{2 + \lambda}{\sqrt{\lambda^2 + 4\lambda + 9}} \Rightarrow \lambda = -4$

$\Rightarrow$ Equation of R: $x + 2y - 2z + 30 = 0$

43. A, C, D

Sol. a_i represent vertices of a regular octagon so

$$\sum_{i=1}^8 \sum_{j=1}^8 \vec{a}_i \cdot \vec{a}_j = 0; \sum_{1 \leq i < j \leq 8} \vec{a}_i \cdot \vec{a}_j = -4$$

$$\sum_{i=1}^8 \sum_{j=1}^8 (\vec{a}_i \times \vec{a}_j) = 0; \sum_{1 \leq i < j \leq 8} \vec{a}_i \times \vec{a}_j = 4(\sqrt{2} + 1)$$

44. A, C

Sol. P(t) is clearly continuous $\forall t$ but non-differentiable whenever the person turns

$$\lim_{t \rightarrow \infty} P(t) = \frac{1}{1 + \frac{9}{16}} \sqrt{1 + \frac{9}{16}} = \frac{4}{5}$$

45. D

Sol. Make cases based on denominator

46. A

Sol. Probability is $\frac{20}{255}$

47. B

Sol. $a = 50, b = 25, c = 50$

48. D

Sol. None of the functions are periodic as local extrema keeps on decreasing after every 4 seconds

Section – B

49. 4

Sol. $2c \sin(A - B) \cos(A + B) = b \sin B - a \sin A$
 $\Rightarrow c[\sin 2A - \sin 2B] = b \sin B - a \sin A$
 $\Rightarrow 2c \sin A \cos A + a \sin A = b \sin B + 2c \sin B \cos B$
 $\Rightarrow \frac{c \cos A + a \cdot \frac{1}{2}}{\sin B} = \frac{b \cdot \frac{1}{2} + c \cos B}{\sin A} \Rightarrow \cos C = \frac{1}{2} \Rightarrow 5 \sin C = \frac{5\sqrt{3}}{2}$

50. 6

Sol. $(\alpha\beta)^3 = \alpha\beta$ and $\alpha^3 + \beta^3 = \alpha + \beta \Rightarrow \alpha\beta = 0, 1, -1$
 If $\alpha\beta = 0 \Rightarrow (\alpha + \beta)^3 = \alpha + \beta \Rightarrow \alpha = \beta = 0, 1, -1$
 Corresponding equation are $x^2 = 0$; $x^2 \pm x = 0$
 If $\alpha\beta = 1 \Rightarrow (\alpha + \beta)^3 = 4(\alpha + \beta) \Rightarrow \alpha + \beta = 0, 2, -2$
 Corresponding equation are $x^2 \pm 2x + 1 = 0$
 If $\alpha\beta = -1 \Rightarrow (\alpha + \beta)^3 + 2(\alpha + \beta) = 0 \Rightarrow \alpha + \beta = 0$
 Corresponding equation is $x^2 - 1 = 0$

51. 8

Sol. $\frac{x^2y^2 + 4 + 4xy + 2y^2 - 2xy^2 - 4y}{xy^2 + 2y} = \frac{(xy + 2)^2 + 2y^2 - 2y(xy + 2)}{y(xy + 2)} = \frac{xy + 2}{y} + \frac{2y}{xy + 2} - 2 \geq 2\sqrt{2} - 2$

Section – C

52. 01.57

Sol. Let $e^x - 1 = \tan^2 y \Rightarrow I_1 = \int_0^{\pi/2} \frac{2\ln(\sec y) - 1}{\ln(\tan y)} dy = \frac{1}{2} \int_0^{\pi/2} \frac{(2\ln \sec y - 1) + (-2\ln(\operatorname{cosec} y) + 1)}{\ln \tan y} dy$
 $= \frac{1}{2} \int_0^{\pi/2} \frac{2\ln \tan y}{\ln \tan y} dy = \frac{\pi}{2}$

53. 00.00

Sol. Let $x = 2 + y \Rightarrow I_2 = \int_{-2}^2 \frac{\ln(2+y)}{\sqrt{4-y^2}} dy = \int_0^2 \frac{\ln(4-y^2)}{\sqrt{4-y^2}} dy$
 Also, let $x = 4 - t^2 \Rightarrow I_2 = 2 \int_0^2 \frac{\ln(4-t^2)}{\sqrt{4-t^2}} dt = \frac{I_2}{2} \Rightarrow I_2 = 0$

54. 01.41

Sol. z_i equals 1, -1, i; $a = \min. \text{ of } \left| z_1 - \frac{kz_2 + (1-k)z_3}{k + (1-k)} \right| = \sqrt{2}$

55. 02.67

Sol. $z = \frac{i}{3} \Rightarrow b = \frac{8}{3}$

56. 01.41

57. 03.16

Sol. (for Q. 56 to 57):

$PA = \sqrt{5}$; $PB = \sqrt{2}$; $PC = \sqrt{10}$